

CS 101 Lab 05 Revision Thu

Q1 Revision — Average Token Length Feature

The toolkit can count tokens but cannot measure text complexity.

Add two new methods:

```
static int countAlphaNumeric(String str)
```

Return the total number of characters that are either letters or digits. You must reuse `isAlphabetic()` and `isNumeric()`.

```
public static double averageTokenLength(String str)
```

Compute:

```
(total letters and digits) / (token count)
```

Requirements:

- must call `countAlphaNumeric()`
- must call existing `tokenCount()`
- if token count is 0 → return 0.0

Menu Update — add a new option:

```
5) Average Token Length  
6) Exit
```

Sample Run:

```
=== String Toolkit ===  
1) Mirror Text  
2) Compare Letters  
3) Token Count  
4) Generate Tag  
5) Average Token Length  
6) Exit  
Select an option (1-6): 5  
Enter text: Hello World  
Average token length: 5.0  
  
Enter text: This is a test  
Average token length: 2.75
```

Q2 Revision — Digit-Aware Shift Cipher

The current `shiftCipher` method shifts only alphabetic characters and leaves digits unchanged. However, in many practical encoding scenarios, digits are also shifted. For example, with a shift of 3:

```
Password123 → Sdvvzrug456
```

The existing implementation does not shift digits.

Modify the existing `shiftCipher(String text, int shift, boolean encode)` **method so that:**

- alphabetic characters are shifted exactly as before
- digits (0–9) are also shifted cyclically

- digit shifting must wrap around (e.g., 9 shifted by 1 becomes 0)
- negative shifts must also work correctly
- non-alphanumeric characters must remain unchanged
- modify the existing loop — do **NOT** write a separate method

Hint: Digit shifting can be implemented similarly to alphabet shifting using modulo 10:

```
newDigit = (originalDigit + shift) % 10
```

Be careful with negative modulo values.

Sample Run:

```
=== Encoding Toolkit ===
1) Shift Cipher
2) Reverse Alphabet Cipher
3) Reverse Text
4) Exit
Choose an option (1-4): 1
Enter text: Password123
Enter shift amount: 3
Type 'e' to encode or 'd' to decode: e
Result: Sdvzrug456

Enter text: Sdvzrug456
Enter shift amount: 3
Type 'e' to encode or 'd' to decode: d
Result: Password123
```